For: ATHE Level 4 Extended Diploma in Computing

Unit 11 Advanced Programming Assignment

Note to Learners

You will need to setup a test FTP server with suitable data files as part of the development process. Your tutor will support you with this. In addition, you may leverage a GUID for each error log entry using an external API, e.g. https://www.uuidtools.com/api/generate/v1

Assignment Scenario

You have been working as a developer for a medium-sized software development studio, headquartered in the large municipality of Centrala, that specialises in building commercial software applications for various platforms.

Centrala's University has a trusted scientific partner that they work with for leading medical research projects. The university's School of Medicine needs to be able to download critical pharmaceutical trial data from this partner in manner that is both secure and reliable.

In the longer term, the scientific partner intends to migrate to a more sophisticated RESTful web-based API, but at the moment they are still using a somewhat simplistic FTP service, despite the accepted security risks and known vulnerabilities this involves.

The University has recently approached your studio to help with the development of a tool that can be used by IT literate staff in the medical school to safely search, download, validate and safely store this data in an interactive manner, e.g., requesting data for a specific date (the default being "today") through command line interface (CLI) or graphical user interface (GUI).

All data must be strictly validated (using a set of rules) before it is accepted, copied for investigation, and then securely archived in a logical directory hierarchy of your creation, although a preference for date and time-based folder naming has been suggested.

Due to the pharmaceutical nature of the data, it is incredibly important that there are no duplicated data sets; this could create bias problems during analysis. Consequently, a tracking mechanism must be designed to ensure each data set is truly unique, i.e. no data file should be downloaded twice.

Sample data – known issues

Unfortunately, previous manually downloaded CSV files have had multiple issues.

This has led Centrala University's School of Medicine to suspect that some data files are being manually collated and edited by their research partner before they are uploaded to the FTP Server. This has led to issues with communication in relation to the variable quality of the data.

Common issues that CU's School of Medicine have identified include:

- Incorrectly formatted filenames
- Duplicated batch_ids in a single file
- Missing headers or misspelt/incorrect headers, e.g. "batch" rather than "batch_id"
- Missing columns on one or more rows
- Invalid entries, e.g., reading values of 10 or greater
- Invalid empty "0 byte" files (because there can be no "nil" returns)
- Malformed files (causing difficulties when importing into other applications, e.g., spreadsheets, databases, cloud-based machine learning etc.)

It is important that any CSV data files which are found to contain any of these issues must be both identified, excluded, and logged using an appropriate technique, e.g. creation of a suitable error log file.

Although it is intended that these logs may one day be incorporated by another application for monitoring and reporting purposes, this is currently outside current development scope.

Sample data – Comma Separated Value (CSV) format

As noted, the University's partner trial data is exported from proprietary medical devices as CSV files, which should follow rfc4180 guidelines, cf. https://tools.ietf.org/html/rfc4180

For example, a valid trial data CSV file, in raw form:

```
"batch_id","timestamp","reading1","reading2","reading3","reading4","reading5","rea
ding6","reading7","reading8","reading9","reading10"
55,"14:01:04",9.875,9.138,1.115,8.006,3.84,4.952,9.038,1.046,2.179,8.701
64,"14:01:04",4.168,9.247,1.958,1.65,3.631,9.317,8.182,9.292,5.978,3.06
116,"14:01:04",2.253,9.489,6.534,8.601,7.397,5.096,8.501,0.541,4.263,8.465
74,"14:01:04",0.89,5.942,3.02,9.526,8.478,7.84,1.85,8.81,4.712,6.805
142,"14:01:04",8.872,7.109,1.453,0.005,8.102,1.598,8.25,5.257,6.01,2.153
129,"14:01:04",2.115,7.738,4.946,7.444,6.362,5.735,9.768,6.25,5.32,4.073
113,"14:01:04",2.25,3.428,2.914,6.452,5.189,2.858,5.935,3.574,7.477,1.61
32,"14:01:04",1.956,6.395,7.196,2.823,2.563,1.305,7.938,6.421,0.398,3.024
172,"14:01:04",2.814,4.611,4.866,9.033,2.711,0.765,7.837,5.17,5.802,0.92
192,"14:01:04",0.539,0.103,1.669,7.162,0.912,7.134,2.151,7.439,6.612,6.731
```

And viewed in Microsoft Excel:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	batch_id	timestam	reading1	reading2	reading3	reading4	reading5	reading6	reading7	reading8	reading9	reading10		
2	55	14:01:04	9.875	9.138	1.115	8.006	3.84	4.952	9.038	1.046	2.179	8.701		
3	64	14:01:04	4.168	9.247	1.958	1.65	3.631	9.317	8.182	9.292	5.978	3.06		
4	116	14:01:04	2.253	9.489	6.534	8.601	7.397	5.096	8.501	0.541	4.263	8.465		
5	74	14:01:04	0.89	5.942	3.02	9.526	8.478	7.84	1.85	8.81	4.712	6.805		
6	142	14:01:04	8.872	7.109	1.453	0.005	8.102	1.598	8.25	5.257	6.01	2.153		
7	129	14:01:04	2.115	7.738	4.946	7.444	6.362	5.735	9.768	6.25	5.32	4.073		
8	113	14:01:04	2.25	3.428	2.914	6.452	5.189	2.858	5.935	3.574	7.477	1.61		
9	32	14:01:04	1.956	6.395	7.196	2.823	2.563	1.305	7.938	6.421	0.398	3.024		
10	172	14:01:04	2.814	4.611	4.866	9.033	2.711	0.765	7.837	5.17	5.802	0.92		
11	192	14:01:04	0.539	0.103	1.669	7.162	0.912	7.134	2.151	7.439	6.612	6.731		
12														
13														

MED_DATA_20230603140104

Filename convention

Filenames from the partner's trial data exports are meant to be created using the following naming convention:

```
MED_DATA_YYYYMMDDHHMMSS.csv
```

For example:

```
MED_DATA_20230603140104.csv
```

This would represent a trial data .csv file that was created at 2:01:04 pm on 03/06/2023.

Batches

Each batch of results is identified by a unique **batch_id** (see 1st field) and this may appear unsorted in *any* .csv file. However, duplication of the same batch_id *within a single file* would indicate a faulty .csv file and this file should be both logged and excluded from further study, i.e. not stored by the University's School of Medicine.

In reality, batches within a single file may be exported within the same second or over multiple seconds, depending on throughput of the scientific partner's proprietary devices, i.e. multiple entries for the same timestamp are not considered unusual.

Data Range

Each and every batch must have 10 readings.

All 10 readings should be represented as floating point numbers, formatted up to three decimal places with no value *exceeding* 9.9 (as this would be considered invalid data).

Remember: You will need to setup a test FTP server with suitable data files as part of the development process.

Task 1

Before starting the development for the CU, Luzo Okake, CEO of the studio has asked you to prepare a briefing document covering the programming paradigms that could be used (singularly or jointly) to solve this problem.

This process will help you to review and select the appropriate paradigm(s) to use for the project.

This document should cover:

- A description of the common programming paradigms that exist and the languages which employ them.

Now that the paradigm(s) and associated programming language have been selected, design and implementation of the solution is ready to begin.

Luzo Okake has recommended you adopt a TDD approach to the software development lifecycle, particularly as the validation of each CSV data file can be broken down into a sequence of discrete validation checks.

Therefore, you should design and implement the following elements:

- A solution which contains designs that pair appropriate algorithms and data structure to achieve efficient processing that will:
    - Search the remote FTP server
    - Download “unseen”/new files
    - Filter them through a sequence of discrete validation checks
    - Store valid files in an appropriate file structure
    - Log invalid files in an error log
- Demonstrate the use of TDD to solve this problem successfully.
- Design a user interface (as requested) which demonstrates the use of recognised principles.

Extension activities:

To achieve a **Merit**, Luko wants you to implement appropriate time-saving techniques which improve maintenance and code reuse:

- Demonstrate the use of a common design pattern as part of a coded solution.

Learning outcomes and assessment criteria

LO1 1.1, 1.2, 1.3, 1.4, 1M1

Task 2

Apply the use of automation techniques to ease the development process.

For example, this could include the use of scripted techniques to generate good/bad sample csv data files for internal testing.

Extension activities:

To achieve a **Merit**, Luko has suggested that you may be able to leverage a GUID for each error log entry using an external API, e.g. https://www.uuidtools.com/api/generate/v1

- Demonstrate the use of an API to perform common tasks needed as part of a software project’s development.

Learning outcomes and assessment criteria

LO2 2.1, 2M1

Task 3

Your software development project must use modern software engineering practice. As such you must be able to:

- Demonstrate the maintenance of a remote repository containing your project's codebase through the use of version control software (i.e. a VCS such as Git, for example).

- Create and deploy a containerised version of your software application.

Extension activities:

To achieve a **Merit**, Luko wants you to:

- Create and deploy a CI/CD pipeline for your software project, including the definition of suitable stages and jobs.

To achieve a **Distinction**, you must:

- Evaluate the contribution of Agile techniques, ceremonies, and asset creation on your software project; did they help or hinder? Provide supporting reasons for your judgements.

Learning outcomes and assessment criteria

LO3 3.1, 3.2, 3M1, 3D1